

Chapter 5 — Functions

AI Study Assistant — RAG Knowledge Base

Learning Objectives

Understand why functions are used.

Define and call functions.

Use parameters and arguments.

Return values from functions.

Understand local variables and basic function design.

1. What is a Function?

A function is a reusable block of code designed to perform a particular task. Functions make programs easier to organize, test, and maintain.

2. Defining a Function

Python functions are defined with the `def` keyword.

```
def greet():  
    print("Hello")
```

The function body is indented.

3. Calling a Function

Defining a function does not execute it. The function runs when it is called.

```
greet()
```

4. Parameters and Arguments

Parameters are names listed in a function definition. Arguments are the values supplied when the function is called.

```
def greet(name):  
    print("Hello", name)  
  
greet("Ayesha")
```

5. Return Values

The `return` statement sends a value back to the code that called the function.

```
def add(a, b):  
    return a + b  
  
result = add(4, 6)
```

Here, `result` receives the returned value 10.

6. Default Parameters

A parameter can have a default value. The default is used when the caller does not provide that argument.

```
def greet(name="Student"):  
    print("Hello", name)
```

7. Scope and Local Variables

A variable created inside a function is generally local to that function. It can be used within the function where it is defined.

8. Why Functions Matter

Functions reduce repetition, improve readability, support modular design, and make it easier to test individual pieces of a program.

Key Points for Revision

def defines a function.

A function call executes a function.

Parameters receive values supplied as arguments.

return sends a value back to the caller.

Functions help make code reusable and modular.